

Car Dealers Sales Analysis – Analytical Process Report

Introduction

This report documents the end-to-end analytical process used in developing the Car Dealers Sales Analysis dashboard. The objective of the analysis was to uncover sales trends, customer demographics, product performance, and revenue insights to support data-driven decision-making for car dealerships. The final output is an interactive dashboard designed for business stakeholders.

Business Objective

- Analyze overall sales performance
- Identify top-performing car models and body styles
- Understand sales trends over time
- Examine regional sales distribution by price and quantity
- Analyze customer demographics (gender)
- Enable dynamic filtering for deeper insights

Data Understanding

The dataset contained car sales records spanning multiple years, with attributes including car model and body style, Transmission type, Sales price and quantity, Customer gender, Sales date, and Dealer region, etc.

The initial exploration focused on understanding the data structure, field definitions, and identifying data quality issues.

Data Cleaning & Preparation

The data cleaning phase ensured accuracy, consistency, and usability:

- Removed duplicate records
- Handled missing values in key columns
- Standardized categorical fields (e.g., transmission type, body style)
- Converted date fields into proper datetime formats
- Created calculated fields such as total sales and average price

This step was critical to ensure reliable analysis and visualization outputs.

Data Transformation

Additional transformations were applied to support analysis:

- Aggregated sales by month for time-series analysis
- Grouped sales by region to compare price and quantity
- Ranked car models to identify top performers
- Created KPI metrics: Total Cars Sold, Average Car Price, Average Annual Income

These transformations enabled efficient slicing and filtering within the dashboard.

Exploratory Data Analysis (EDA)

EDA was conducted to identify patterns and trends:

- Sales growth trends over time
- Comparison of revenue versus quantity across regions
- Identification of the most popular car models
- Customer gender distribution analysis

Insights from this phase informed dashboard design and KPI selection.

Key Insights Summary and Recommendations

1. Car sales show a **gradual increase throughout the year**, with a **notable spike towards the last quarter**, especially around **October–December**. This suggests a strong **seasonal buying pattern**.

Recommendations

- Dealerships should **increase inventory and marketing spend in Q4**
 - Introduce **year-end promotions, financing offers, and bundle deals**
 - Plan staffing and logistics ahead of peak months to avoid stock-outs
2. A small group of car models dominates sales volume, indicating **customer preference concentration** rather than evenly distributed demand across all models.

Recommendation

- Prioritize **high-demand models** in inventory planning
 - Reduce overstock of low-performing models
 - Use top-selling models as **entry products** to upsell higher-margin vehicles
3. Some regions record **high sales quantity but lower total revenue**, while others show **lower volume but higher average price**, indicating **distinct regional purchasing power**.

Recommendation

- Apply **region-specific pricing strategies**
- Promote premium models in high-income regions
- Focus on affordable, high-volume models in price-sensitive regions

4. Male customers account for a **significantly higher proportion of purchases** compared to female customers.

Recommendation

- Develop **targeted campaigns** aimed at female buyers
- Highlight safety, fuel efficiency, and family-oriented features
- Partner with lifestyle brands or offer test-drive events tailored to female audiences

5. Automatic transmission and specific body styles (e.g., SUVs and Sedans) show **stronger sales performance**, indicating a preference for comfort and versatility.

Recommendation

- Increase the stocking of automatic vehicles
- Promote SUVs and Sedans in advertising campaigns
- Use manuals strategically for budget-conscious buyers

Conclusion

This project demonstrates a complete data analytics workflow from raw data cleaning to actionable business insights. The dashboard provides stakeholders with an intuitive tool to monitor performance and make informed decisions. This analysis highlights strong proficiency in data preparation, analysis, visualization, and business storytelling.